Practical - 1

Practical Title: Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

Problem Definition 1: Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and minimum 10 pages. Create a sitemap, low-fidelity wireframe, project folder structure, README file, and GitHub repository.

A. Theory

1. What is URL and parts of URL?

Answer: URL means (Uniform Resource Locator) . It is the address of something on the internet. It have parts like:

1)Protocol: https or http

2)Domain name: Host's Name

3)Port Number: 443,200,etc

4)Path: Specific address to something in the internet.

I used the help of Google Gemini and ChatGpt.

2. How does HTML file process in web browser?

Answer:

User Opens URL / HTML File



Browser Sends Request



Server Returns HTML File



Browser Parses HTML



Creates DOM



Loads CSS & JavaScript

▼

Renders and Displays Webpage

References: Duckett, J. (2021). *HTML and CSS: Design and Build Websites*. John Wiley & Sons.

3. How will page navigation flow managed among all HTML pages?

Answer: A sitemap is an important part of a website because it improves navigation for users and helps search engines understand the website's structure. This makes the website more user-friendly and improves its visibility in search engine results.

Uses of a Sitemap

1. **Improves Website Navigation**
   - A sitemap helps visitors quickly locate the information or page they are looking for.
2. **Helps Search Engines**
   - Search engines use sitemaps to find and index website pages, making them easier to appear in search results.
3. **Organizes Website Structure**
   - It provides a clear overview of all the pages and their relationships within the website.
4. **Speeds Up Indexing**
   - Newly added or updated pages can be discovered and indexed faster by search engines.
5. **Useful for Large Websites**
   - Websites with many pages benefit from sitemaps because they make content easier to manage and access.

References:

Duckett, J. (2021). *HTML and CSS: Design and Build Websites*. John Wiley & Sons.

4. How will GitHub commits be maintained after each practical?

Answer: GitHub commits should be maintained by making a commit after completing each practical. Every commit should include only the changes made for that particular practical and have a clear, meaningful commit message describing what was done. This creates a record of the work completed over time and makes it easy to track progress.

Example Commit Messages

- Added Practical 1 - HTML Basics
- Completed Practical 2 - Lists and Tables
- Updated Practical 3 - Forms

- Fixed errors in Practical 4

B. Evidence of Work (Screenshots):

```
REQUIREMENT ANALYSIS:
login page
home page-attendance, contact, FAQ, about us,
fees-scholarships
exam-timetable,result
faculty list-what they teach
leave application
paper assessment
hall ticket
lecture material
feedback and complaints
```

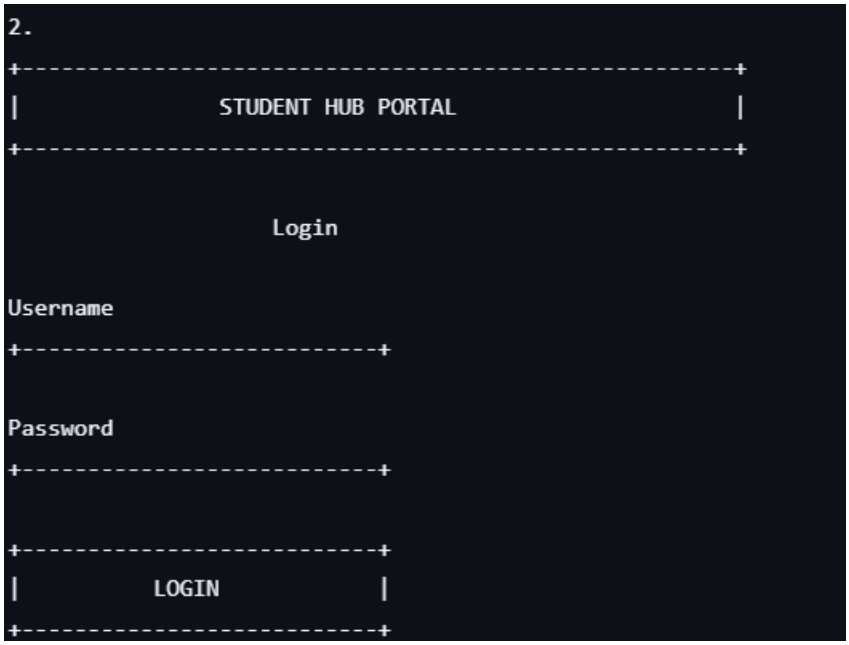
```
SITEMAP:
THIS IS THE SCRIPT MAP OF THE STUDENT HUB PORTAL I WANT TO MAKE:
IT IS NOT PERFRECT WEBSITE BUT I THINK THIS MANY FEATURES ARE ENOUGH FOR A STUDENT FOR A GOOD EXPERIENCE.
------LOGIN PAGE-------->LOGIN OR SIGN UP-------->HOME PAGE------>1)ATTENDANCE,2)HALL TICKET,3)LEAVE APPLICATION,4)PAPER ASSSESMENT
                                                                |          |               |               |
                                                                          --->HALL TICKET     --->LEAVESS       --->KNOW YOUR ANSWERSHEET
                                                                ->%ATTENDANCE
------>FEES---->TIMETABLE--->RESULT---->CONTACT
```

```
1.
+-----------------------------------------------------+
|                  STUDENT HUB PORTAL                 |
+-----------------------------------------------------+

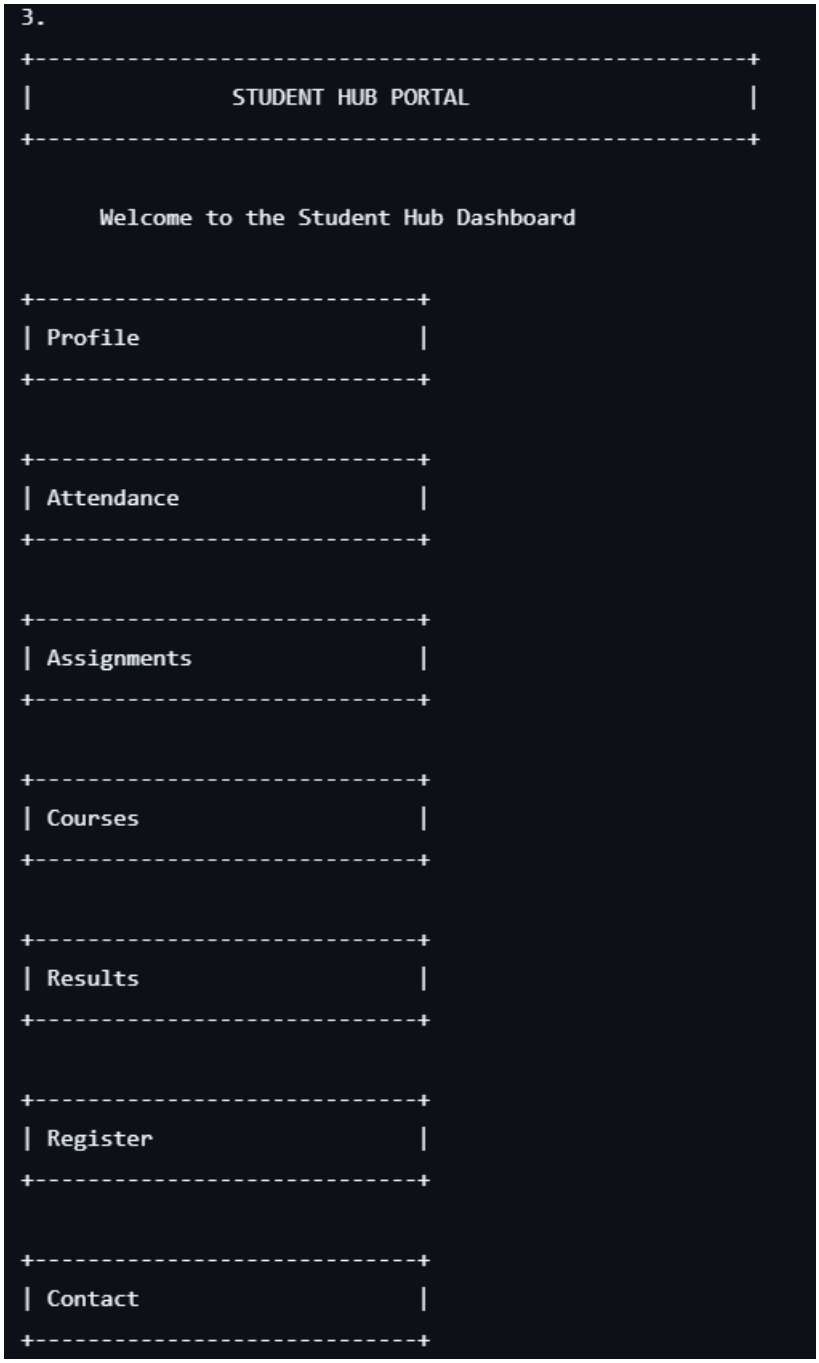
          Welcome to the Student Hub Portal!

  This website provides a centralized platform for
  students to access various resources and services.

                [ Login Button ]
```


2.
STUDENT HUB PORTAL
Login
Username
Password
LOGIN

3.


STUDENT HUB PORTAL
Welcome to the Student Hub Dashboard
Profile
Attendance
Assignments
Courses
Results
Register
Contact

4.

```
+-----------------------------------------------------+
|                STUDENT PROFILE                      |
+-----------------------------------------------------+

              [ Profile Photo ]

Name        : ____________

Roll No     : ____________

Email       : ____________

Phone       : ____________

Address
_____________________________________

_____________________________________
```

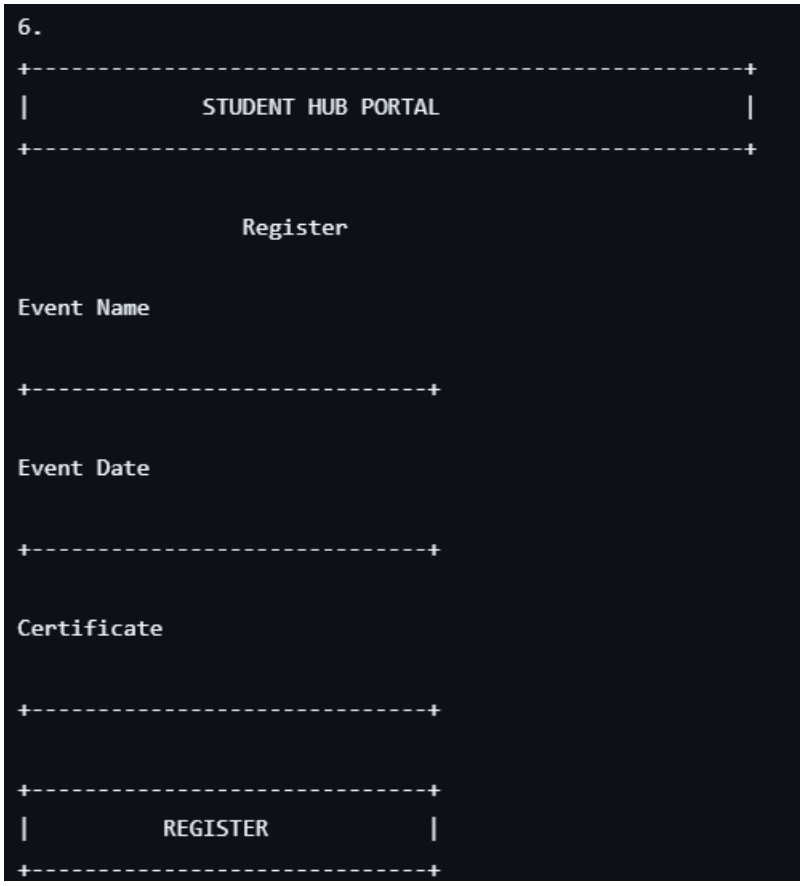
5.

```
+-----------------------------------------------------+
|              STUDENT HUB PORTAL                     |
+-----------------------------------------------------+

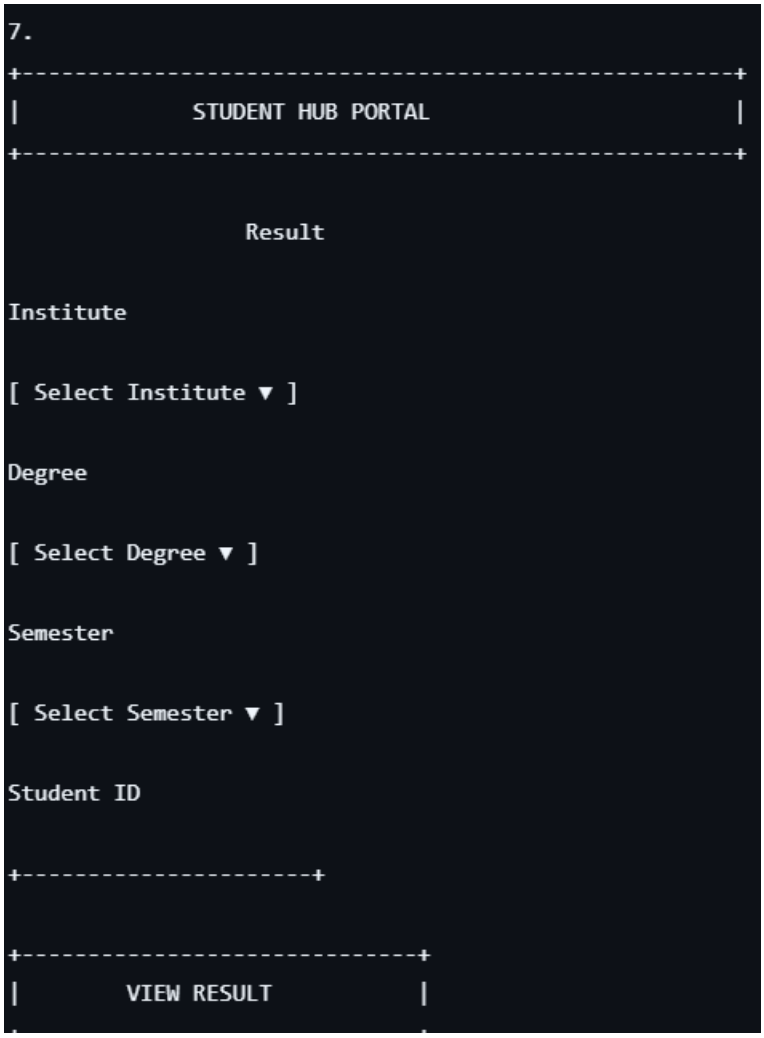
                Attendance

Overall Attendance : 80%

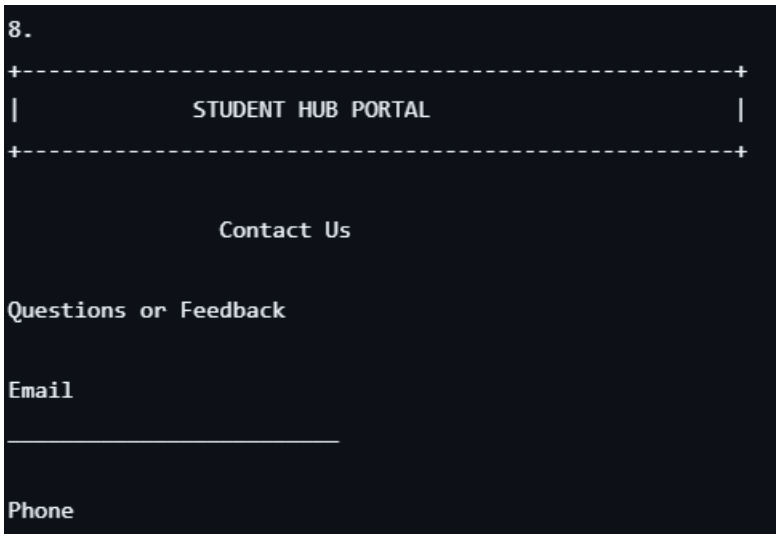
-------------------------------------------------------
| Subject | Lectures Attended | Attendance Percentage  |
-------------------------------------------------------
| MSUD    |      12/15        |        80%             |
-------------------------------------------------------
| CEUC    |      10/15        |      66.67%            |
-------------------------------------------------------
| DSA     |      14/15        |      93.33%            |
-------------------------------------------------------
| WDF     |      13/15        |      86.67%            |
-------------------------------------------------------
| HSUV    |      11/15        |      73.33%            |
-------------------------------------------------------
```


6.
STUDENT HUB PORTAL
Register
Event Name
Event Date
Certificate
REGISTER

7.


STUDENT HUB PORTAL
Result
Institute
[Select Institute ▼]
Degree
[Select Degree ▼]
Semester
[Select Semester ▼]
Student ID
VIEW RESULT

8.

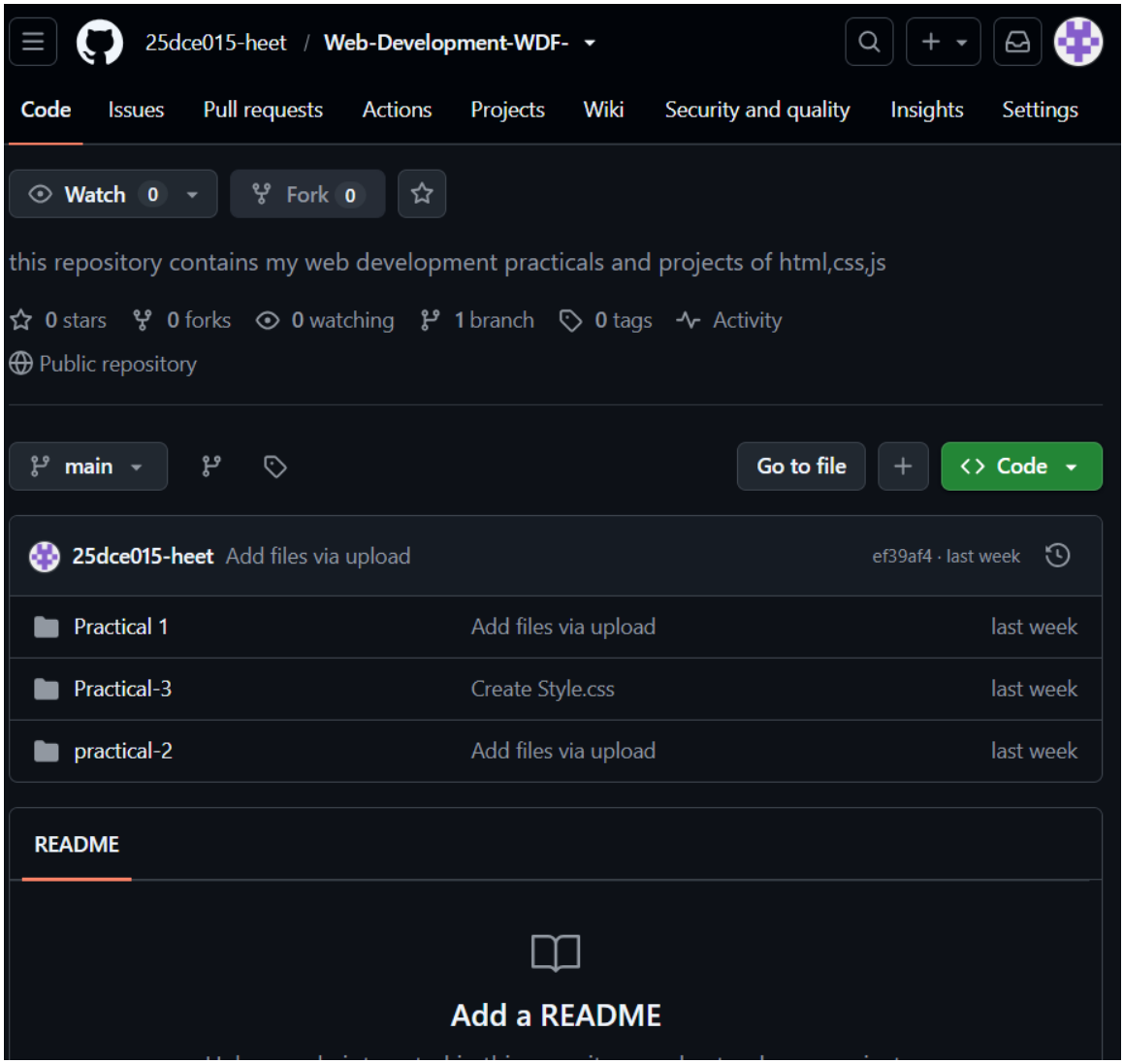

STUDENT HUB PORTAL
Contact Us
Questions or Feedback
Email
Phone

C. GitHub Repository Link:

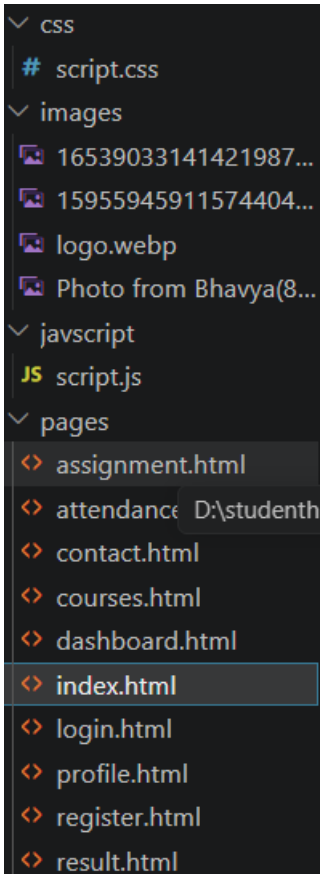
https://github.com/25dce025-debug/WEB-DEVELOPMENT-WDF-

D. Challenges Faced:

GITHUB REPO. CREATION,FOLDER CREATION,FILE CREATION.



VSCODE FOLDER,FILE CREATION.



E. Learning Outcomes:

I LEARNED WEBSITE CREATION, SITEMAP, WIREFRAME, GITHUB, REQUIREMENT ANALYSIS, FILE MANAGEMENT

Faculty Remarks: (Keep this blank for faculty)